Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

Win condition: Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

THE SCIENCE YOU ARE USING

Drag, lift, and hang time. A maple samara spins as it falls. The wing generates lift that slows the descent, so even a light breeze carries the seed away from the shade of its parent. More surface area for the same mass usually means a longer, slower fall.

One variable at a time. Wing length, wing angle, and added mass all change flight. Engineers learn what each does by changing only one and comparing. Copying a natural strategy on purpose is biomimicry.

HOW TO BUILD AND RUN IT

1. **Study a real samara.** Drop a maple seed or a model. Watch how the wing makes it spin and slow down.
2. **Build a baseline seed.** Make a simple winged seed with one paper clip for mass. Record a baseline drop time and distance.
3. **Change one variable.** Adjust wing length OR mass OR fold angle, only one. Predict the effect before testing.
4. **Test for hang time and distance.** Drop from the standard height or release in the fan lane. Time three trials and average.
5. **Aim for a target.** In a final round, try to land in a marked zone. Control plus distance both score.

Safety: No throwing at people. Use marked launch lanes and low-power fan settings. Allergy: None.

MATERIALS

Paper and cardstock	shared
Paper clips	for mass
Masking tape	shared
String	shared
Fan or marked lane	1 test zone
Timers	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**